DA-23 Salary Band & Pay Gap Analysis

Step 4: Exploratory Data Analysis (Python)

Business question: Is pay consistent across roles and gender?

Data: 10,500 employees (cleaned dataset from Step 2). Salaries are annual CTC in rupees; 1 lakh = 100,000. Analysis done in Python using pandas, numpy, matplotlib and seaborn.

1. Data Overview

The dataset has 10,500 rows and 16 columns, with no duplicate rows or duplicate employee IDs. The only missing values are 210 Joining_Date entries, which are not used in the analysis. Gender comparisons use the 10,179 employees recorded as Male or Female.

2. Summary Statistics

Variable	Mean	Std Dev	Min	25%	Median	75%	Max
Age	29.7	5.9	19.0	26.0	29.0	33.0	56.0
Experience Years	6.8	5.4	0.0	2.8	5.5	9.2	29.0
Annual Salary INR	1,375,902	1,205,210	156,700	658,725	992,950	1,610,800	11,373,800
Performance Rating	3.3	0.9	1.0	3.0	3.0	4.0	5.0

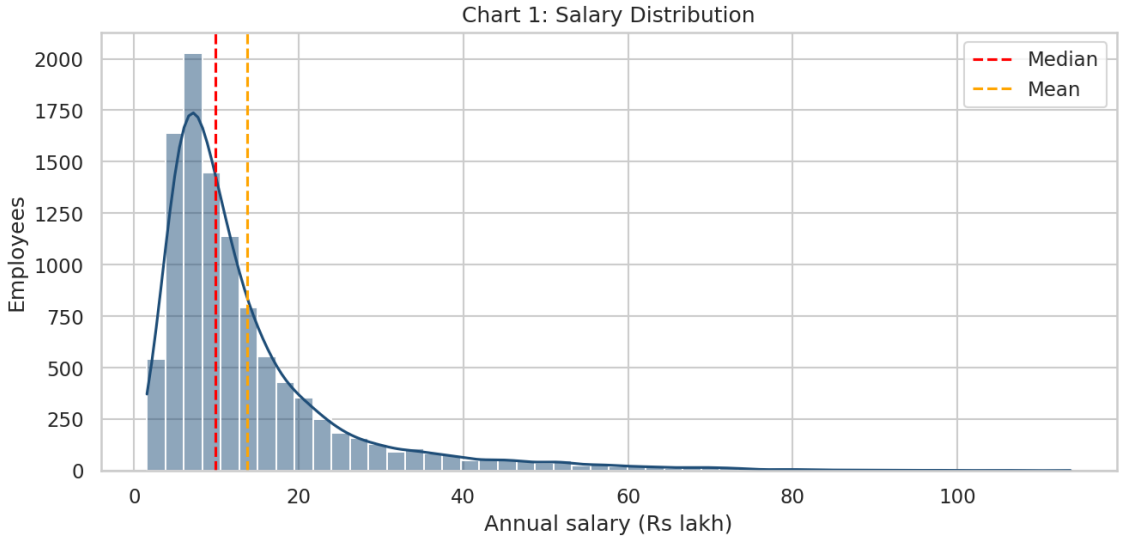
Item	Distribution
Gender	Male 59.4%, Female 37.6%, Not Specified 3.1%
Departments (share)	IT 21.9%, Sales 17.4%, Operations 14.8%, Finance 11.3%, Marketing 9.4%, others below 9.5%
Job roles (largest)	Associate 25.5%, Junior Associate 22.5%, Senior Associate 20.4%

Observation: The average employee is 29.7 years old with 6.8 years of experience. Average salary (Rs 13.76 lakh) is well above the median (Rs 9.93 lakh) and skewness is 2.61, so salary is right-skewed and the **median is the better measure of a typical salary**. IT (21.9%) and Sales (17.4%) are the largest departments.

3. Visual Analysis

Chart 1: How is salary distributed?

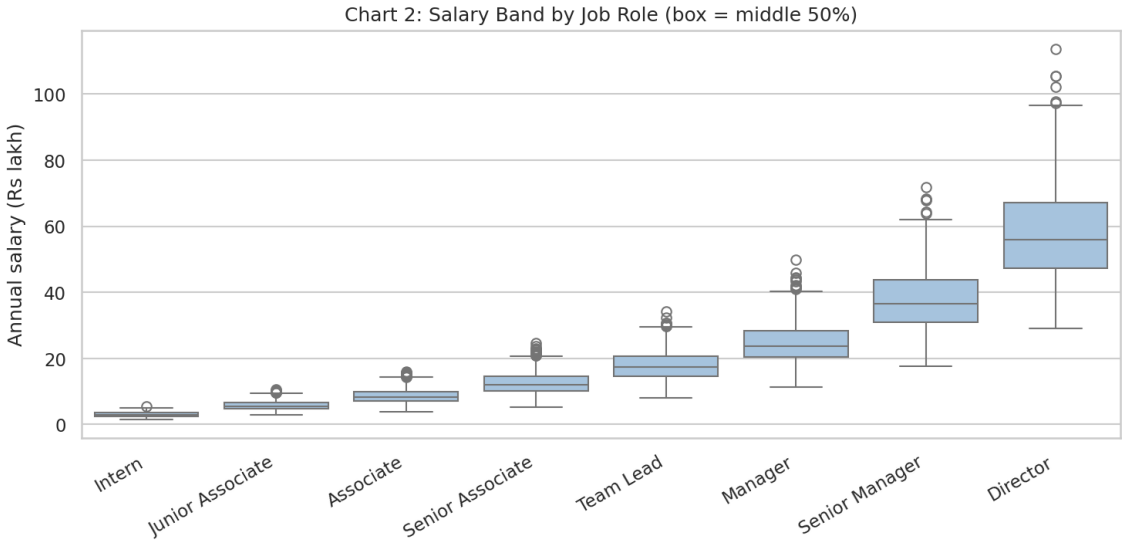
Question: What does a typical salary look like, and is it skewed?
Why this chart: A histogram with KDE shows the shape of one numeric variable; mean and median lines show the skew.



Observation: Most employees earn between Rs 4 and 15 lakh, with a long tail of senior staff above Rs 40 lakh. The mean (orange) is pulled about Rs 3.8 lakh above the median (red) by these high earners.

Chart 2: Salary band by job role

Question: What is the salary band for each role, and are there outliers?
Why this chart: A box plot compares the spread, median and outliers of one number across many categories.

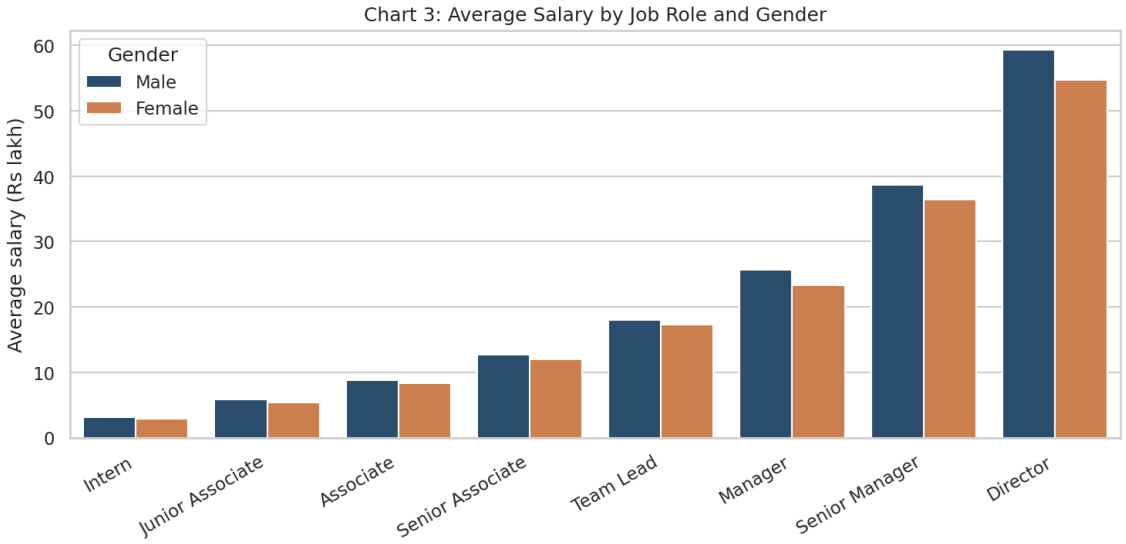


Observation: Salary rises steeply with level, and the spread inside each role widens at senior levels (Director middle 50% is roughly Rs 47 to 67 lakh). Only 81 employees (0.8%) are IQR outliers. They are genuine high earners in their roles, so they are kept.

Chart 3: Average salary by role and gender

Question: Do men and women earn the same in the same role?

Why this chart: A grouped bar chart compares two groups side by side within each category.



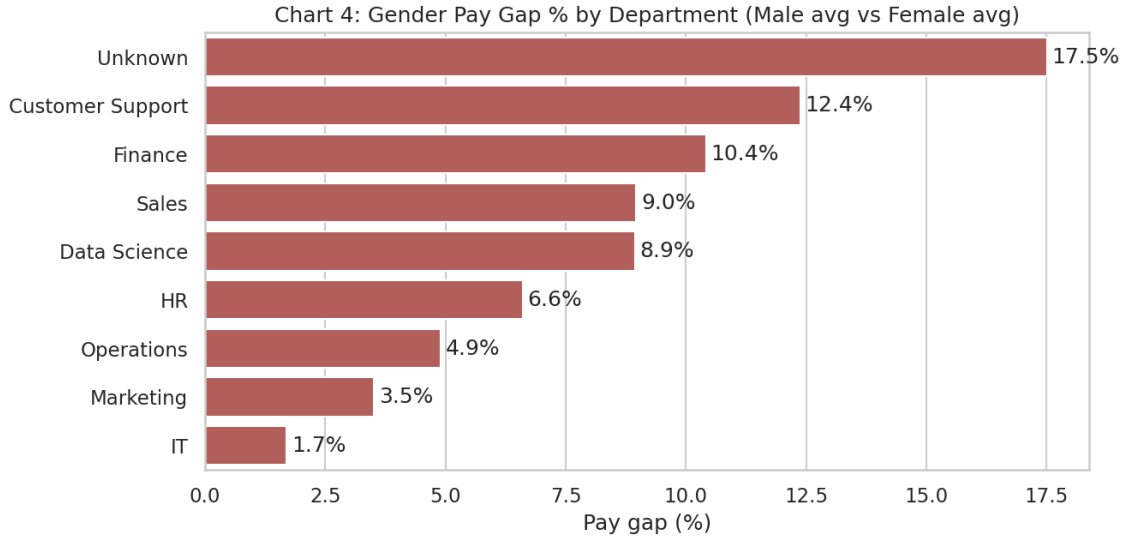
Job role	Male avg (lakh)	Female avg (lakh)	Pay gap
Intern	3.11	2.94	5.38%
Junior Associate	5.82	5.43	6.82%
Associate	8.76	8.29	5.42%
Senior Associate	12.70	11.97	5.78%
Team Lead	18.04	17.29	4.15%
Manager	25.64	23.36	8.89%
Senior Manager	38.69	36.47	5.73%
Director	59.34	54.67	7.87%

Observation: Men earn more than women in **every one of the 8 roles**. The overall gap is **6.24%** (Rs 14.15 lakh vs Rs 13.26 lakh). Within roles it ranges from 4.15% (Team Lead) to 8.89% (Manager), so the gap is not just because women hold junior roles.

Chart 4: Pay gap by department

Question: Which departments have the biggest gender pay gap?

Why this chart: A sorted horizontal bar chart ranks categories and keeps long labels readable.

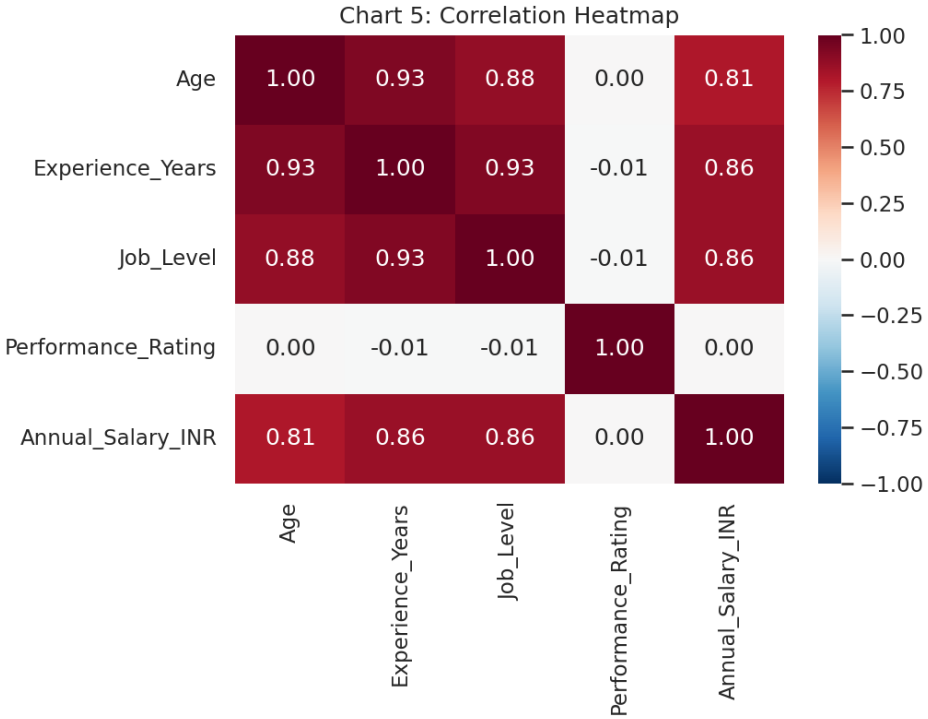


Observation: Customer Support (12.4%) and Finance (10.4%) have the largest gaps among named departments, while IT has the smallest (1.7%). The Unknown group (17.5%) has only 98 employees, so read it with caution, but it shows why department data must be completed.

Chart 5: What drives salary? (correlation)

Question: Which numeric factors move with salary?

Why this chart: A correlation heatmap shows relationships between many numeric columns at once.

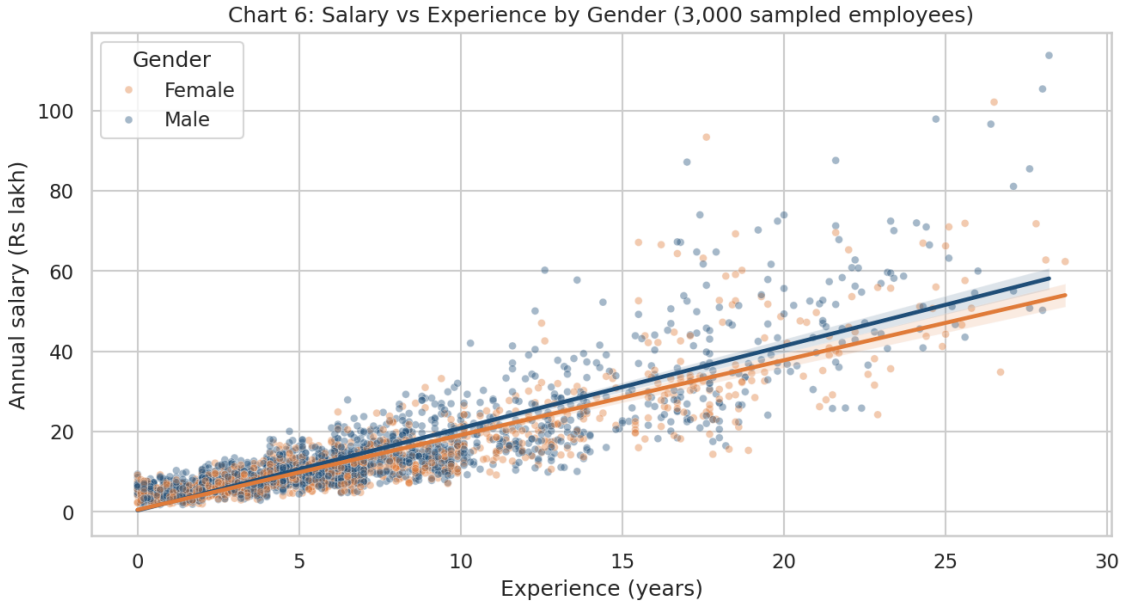


Observation: Salary is strongly linked to experience (0.86), job level (0.86) and age (0.81). **Performance rating has zero correlation with salary (0.00)**, which means pay is not rewarding performance at all. This is a key policy gap.

Chart 6: Salary vs experience by gender

Question: For the same experience, do men and women earn the same?

Why this chart: A scatter plot with trend lines shows the relationship between two numbers, split by group.

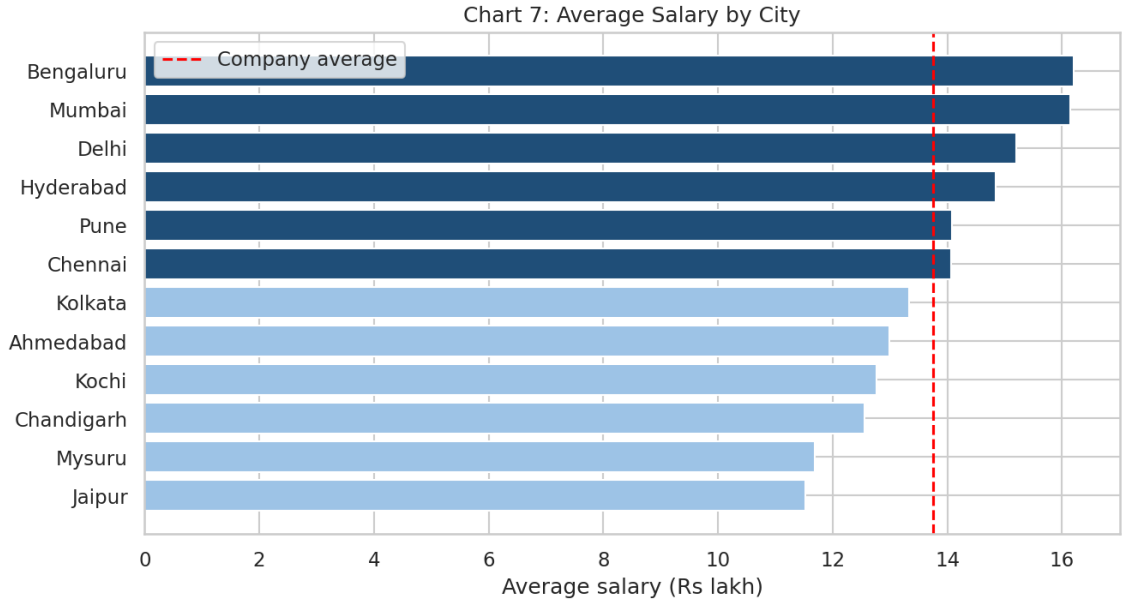


Observation: The female trend line sits below the male line at every experience level, and the rupee gap widens with experience. For equal experience, women are paid less.

Chart 7: Average salary by city

Question: Which cities pay above or below the company average?

Why this chart: A sorted horizontal bar chart with an average line makes ranking and comparison easy.



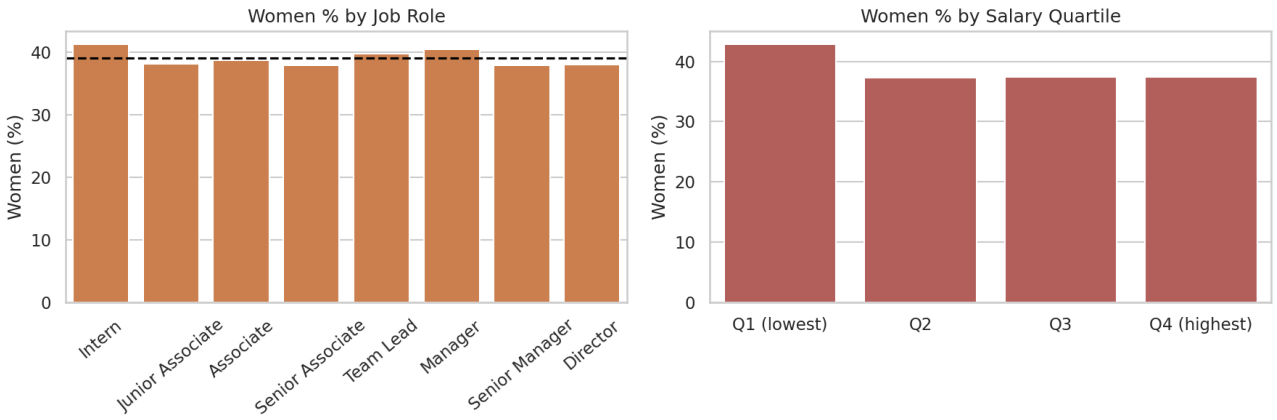
Observation: Bengaluru (Rs 16.20 lakh) and Mumbai (Rs 16.14 lakh) pay the most; Jaipur (Rs 11.53 lakh) pays about 29% less than Bengaluru. Six of the 12 cities are above the company average.

Chart 8: Are women represented fairly?

Question: Are women under-represented in senior roles or in the top pay bracket?

Why this chart: Side-by-side percentage bars compare shares across two different groupings.

Chart 8: Representation of Women



Observation: Women are about 38% to 41% of every job role, so representation by level is balanced. However, women are **42.8% of the lowest-paid quartile and only 37.4% of the top quartile**. The problem is pay within roles, not who gets promoted.

4. Segmentation

4.1 Experience band and gender (avg salary, Rs lakh)

Experience	Male	Female	Gap
0-2 yrs	5.15	4.74	8.1%
2-5 yrs	8.08	7.60	5.9%
5-10 yrs	12.98	12.20	6.1%
10-20 yrs	27.62	26.23	5.0%
20+ yrs	53.18	47.97	9.8%

Observation: The gap exists in every experience band (5% to 9.8%) and is largest for employees with 20+ years of experience (9.8%).

4.2 Performance rating and gender (avg salary, Rs lakh)

Rating	Male	Female	Gap
1	13.76	12.42	9.7%
2	14.20	14.02	1.3%
3	14.29	13.16	7.9%
4	13.78	13.24	3.9%
5	14.52	13.36	8.0%

Observation: Women with a rating of 5 earn 8.0% less than men with a rating of 5. A higher rating does not close the gap, and average pay barely changes between ratings 1 and 5 for either gender.

4.3 Education (salary, Rs lakh)

Education	Employees	Mean	Median
B.E./B.Tech	1,317	15.14	10.62
MCA	1,054	14.09	10.46
MBA	1,014	14.08	10.28
M.Sc	1,017	13.77	10.04
M.Tech	1,261	13.75	9.64
B.Sc	1,239	13.56	9.73
PhD	1,157	13.47	9.30
BBA	1,135	13.00	9.45
B.Com	1,306	12.96	9.85

Observation: B.E./B.Tech has the highest average (Rs 15.14 lakh), but medians for all degrees sit between Rs 9.3 and 10.6 lakh. Education has only a small effect on pay compared with level and experience.

5. Does Anything Explain the Pay Gap?

A regression of log salary on gender, while controlling for job level, experience, performance rating, department and city, was used to test whether other factors explain the gap.

Measure	Result
Raw gap (average, men vs women)	6.24%
Gap after controlling for all factors above	6.48%
Statistical strength (t-statistic)	-31.4 (highly significant)
Model fit (R-squared)	0.977

Observation: After controlling for level, experience, performance, department and city, women still earn about **6.5% less**. The model explains 97.7% of salary variation, so the remaining gap is not explained by any factor in the data.

6. Key Findings

• **Gender pay gap of 6.24%** overall, and 4% to 9% in every role, so it is not a role-mix effect.

• **The gap is about 6.5% even after controlling** for level, experience, performance, department and city.

• **Pay is not linked to performance** (correlation 0.00); performance ratings have no effect on salary.

• **Salary is driven by level, experience and age** (correlations 0.81 to 0.86).

• **Departments differ:** Customer Support and Finance have the widest gaps, IT the narrowest (1.7%).

• **Geography matters:** Bengaluru pays about 29% more than Jaipur.

• **Women are 42.8% of the lowest pay quartile vs 37.4% of the highest**, though representation by role is balanced (38% to 41%).

• **Next step:** Step 5, the Power BI dashboard, to show these KPIs interactively.